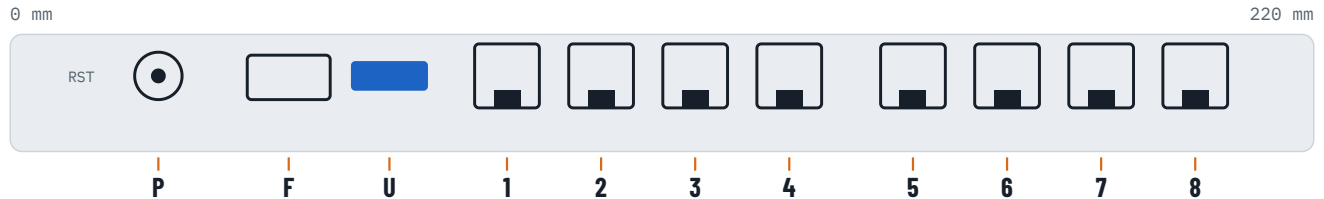


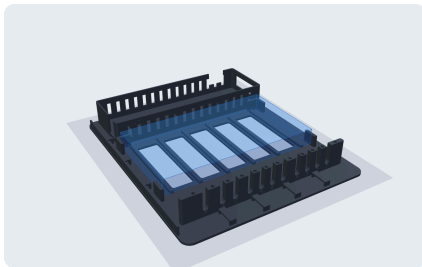
# RB5009 Cable Kit



Six 3D-printable designs built around your router: a 220 × 125 mm RB5009 with eight Ethernet ports, an SFP+ cage, USB and a 24 V DC jack on the front. Every part has been sliced in OrcaSlicer with your printer's profile and passed the safety check, so the times and weights are real estimates.

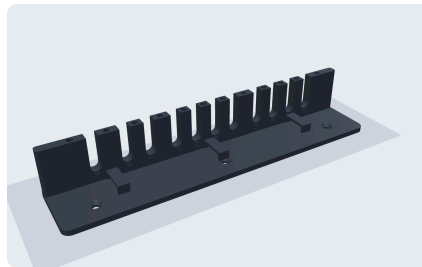


Router front panel to scale. Every comb has one slot per connector with the same label: P = DC power, F = SFP+, U = USB, 1-8 = Ethernet ports.



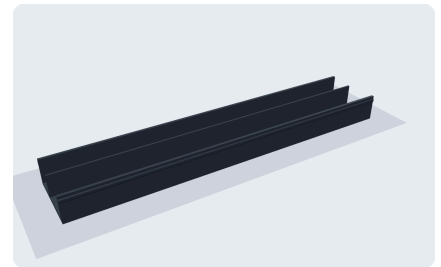
## A Router Dock

7 h 16 min · 236 g



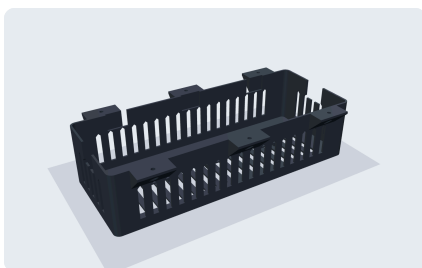
## B Port Comb

2 h 33 min · 64 g



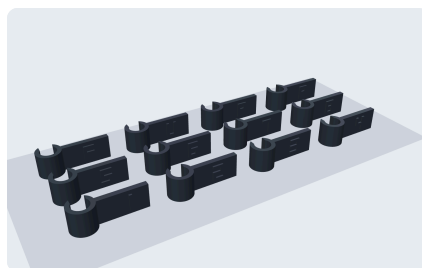
## C Twin-Lane Raceway

3 h 03 min per 300 mm · 127 g



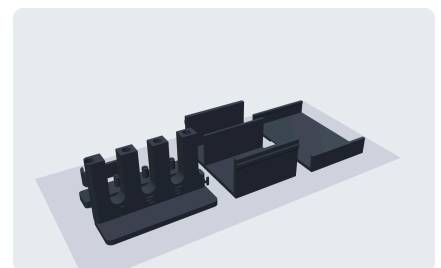
## D Under-Desk Basket

8 h 27 min · 237 g



## E Tags and Holders

1 h 35 min · 32 g



## F Fit Test

1 h 14 min · 29 g

### Start with F

The fit-test plate takes 1¼ hours. Check the comb slots against your ports and feel the raceway lid click, before a long print.

### Best single print: A

The dock organises everything around the router. Overnight, about 7¼ hours.

### Filament

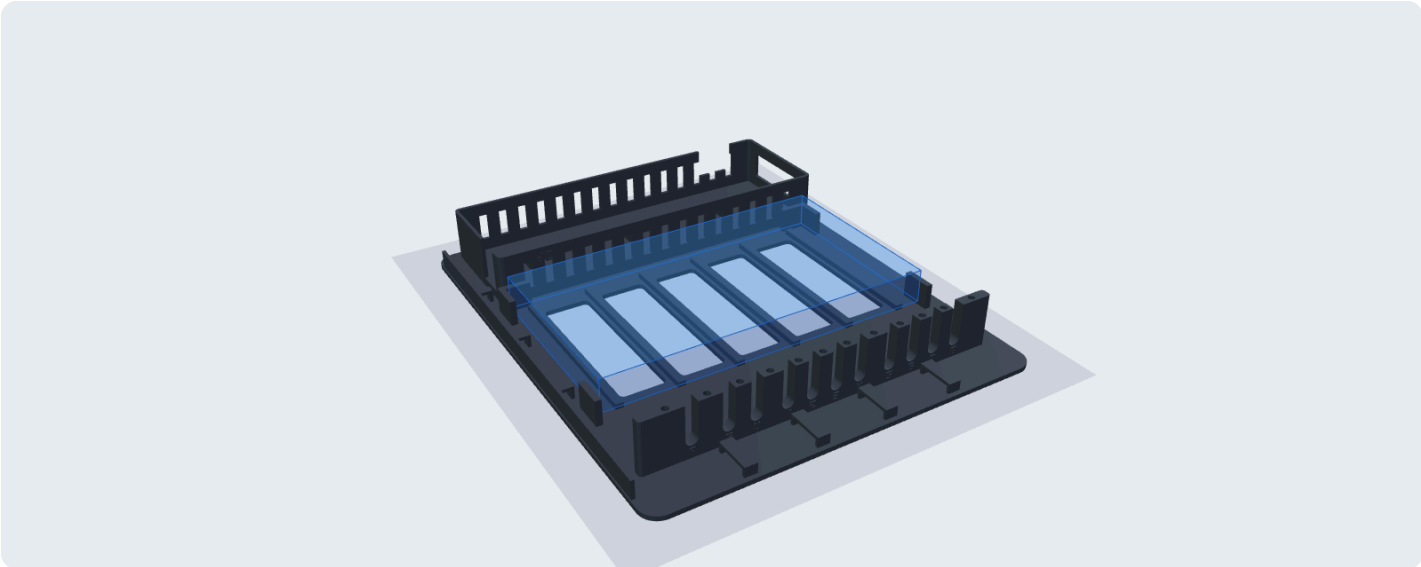
One 1 kg roll, about 985 g left. F + A + E use about 300 g. Everything here, one of each: about 805 g.



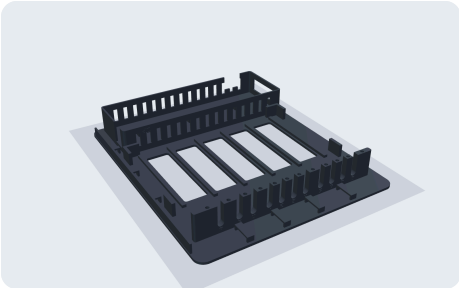
# Router Dock

RECOMMENDED

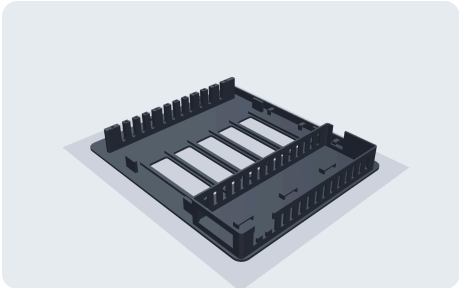
A base plate the router sits in, with everything around it organised: a numbered comb in front of the ports, a separate lane for the DC cord and a vented bay at the back for the power brick.



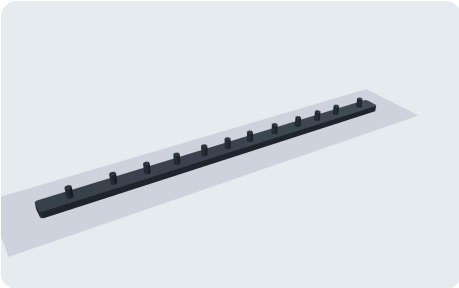
Dock with the router in place (blue outline)



Dock without the router



From the back: brick bay



A2 lock bar (pins up)

## What it does

- Router sits on rails with a 3.5 mm air gap, held by four corner posts, and lifts straight out.
- Comb 45 mm in front of the ports, one slot per connector, labelled P F U 1–8. The lock bar closes the slots.
- DC cord runs from the brick bay along the left lane under three cable bridges to the jack.
- Brick bay: open top, vented walls, three strap loops, cord exits to the lane and out the back.
- Four strap loops in front of the comb to bundle the cables as they leave.

## How to use it

1. Print A1 overnight, then A2 (25 min).
2. Put the dock where the router lives and set the router between the four corner posts, ports facing the comb.
3. Plug in each cable and lay it in the slot with the same number. Lay the power cord in slot P.
4. Press the lock bar onto the top of the comb: its pins go into the holes between the slots.
5. Put the power brick in the back bay and hold it with a velcro strap through the loops. Run the DC cord out of the bay's left opening, forward along the left lane under the three bridges, to the jack.
6. Bundle the cables in front of the comb with velcro straps through the front loops.

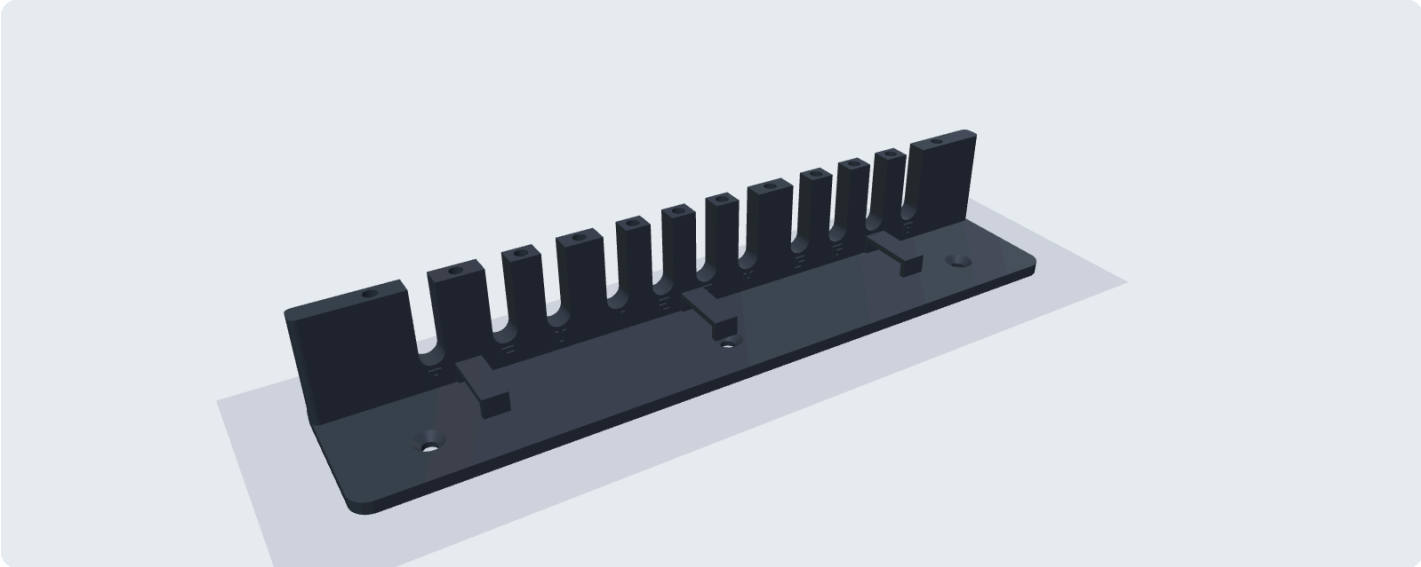
| PART        | SIZE, MM       | PRINT TIME | FILAMENT |
|-------------|----------------|------------|----------|
| A1 dock     | 254 × 300 × 37 | 6 h 51 min | 228.3 g  |
| A2 lock bar | 230 × 11 × 10  | 25 min     | 7.8 g    |
| Total       |                | 7 h 16 min | 236.1 g  |



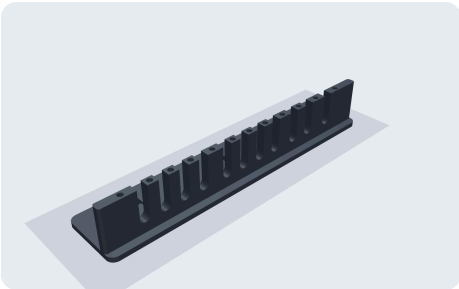
# Port Comb

SIMPLE

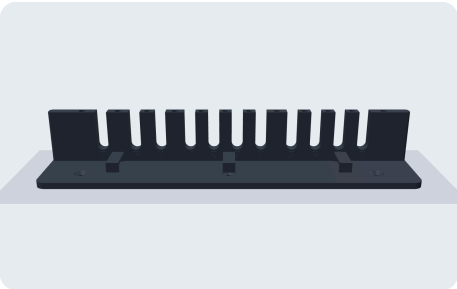
Just the comb, standing on its own foot in front of the router. Good if the router should stay where it is.



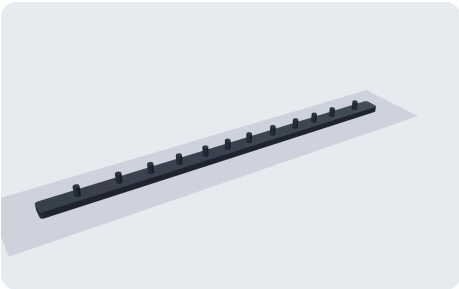
B1 comb



From behind: strap loops



Front: slot labels



B2 lock bar

## What it does

- Same slots and labels as the dock, plus the lock bar.
- Foot with three countersunk screw holes and two recesses for tape pads.
- Three strap loops behind the slots.

## How to use it

1. Print B1 and B2.
2. Stand the comb about 4.5 cm in front of the router's ports. Screw it down (three #8 or M4 countersunk screws) or use foam tape.
3. Lay each cable in its numbered slot and press the lock bar on top.
4. Bundle the cables behind the comb with velcro straps through the loops.

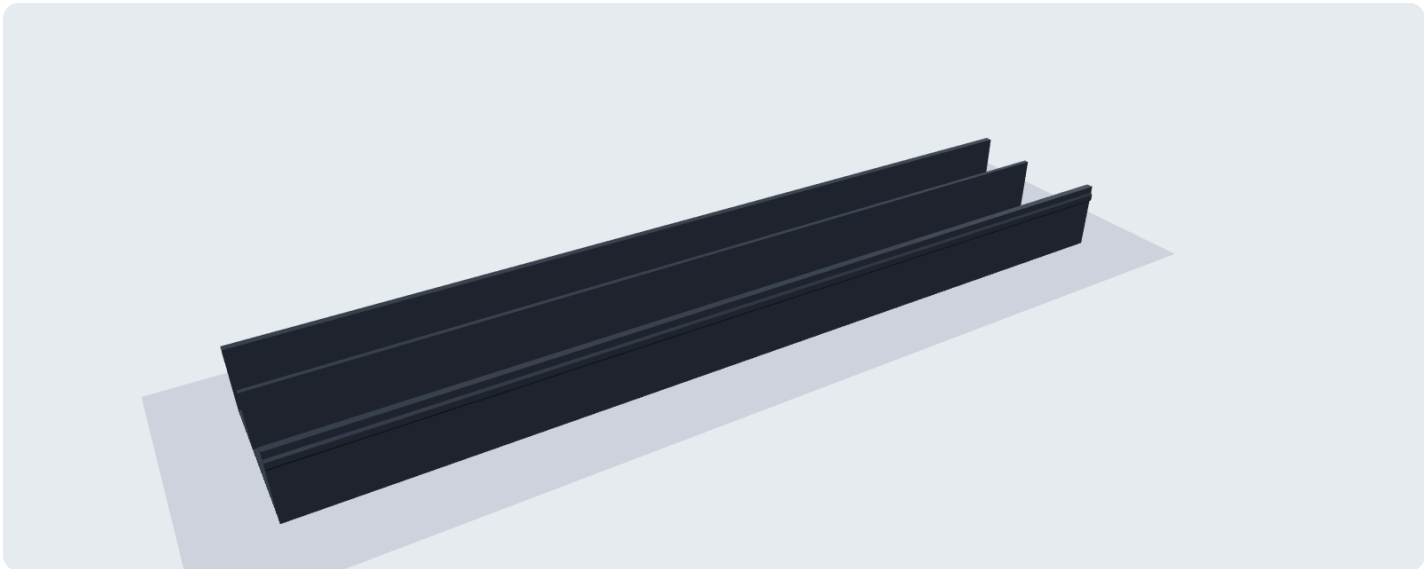
| PART        | SIZE, MM      | PRINT TIME | FILAMENT |
|-------------|---------------|------------|----------|
| B1 comb     | 240 × 51 × 38 | 2 h 07 min | 55.9 g   |
| B2 lock bar | 240 × 11 × 10 | 26 min     | 8.1 g    |
| Total       |               | 2 h 33 min | 64.0 g   |



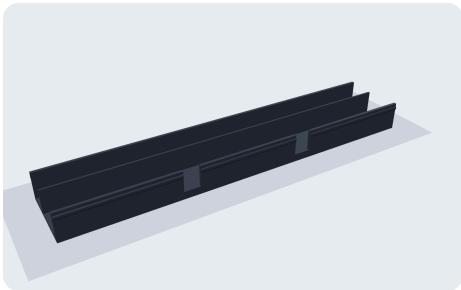
# Twin-Lane Raceway

MODULAR

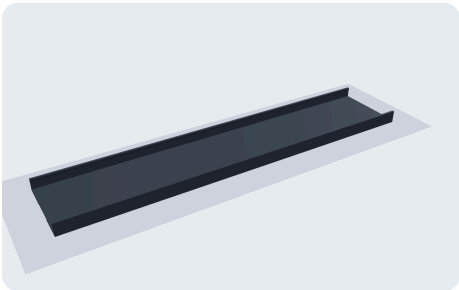
A covered channel for cable runs along a wall, desk edge or skirting board. The data lane and the power lane are separated by a wall, and the lid clicks on.



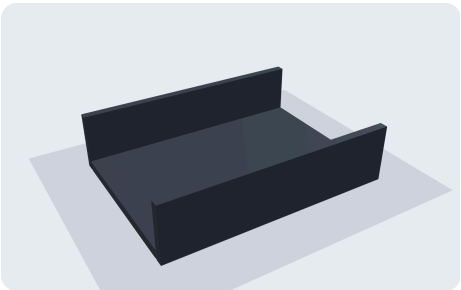
C1 base



C2 base with side exits



C3 lid (printed upside down)



C4 joiner

## What it does

- LAN lane 30 mm wide (about 10–12 Cat6 cables); power lane 19.6 mm (two cords).
- Segments are 300 mm. Fit lids offset from the bases so every seam is covered; the joiner clip lines up two bases.
- C2 has two side exits for cables that leave part way along.
- Countersunk screw holes in both lanes, or mounting tape on the flat back.

## How to use it

1. Plan the run and count 30 cm pieces. Use C2 where cables leave part way along.
2. Screw or tape the bases end to end. A C4 joiner in the LAN lane lines up each seam.
3. Lay network cables in the wide LAN lane and power cords in the narrow lane.
4. Click the lids on, shifted half a piece from the bases so every seam is covered. Pry a lid off at its finger notch.

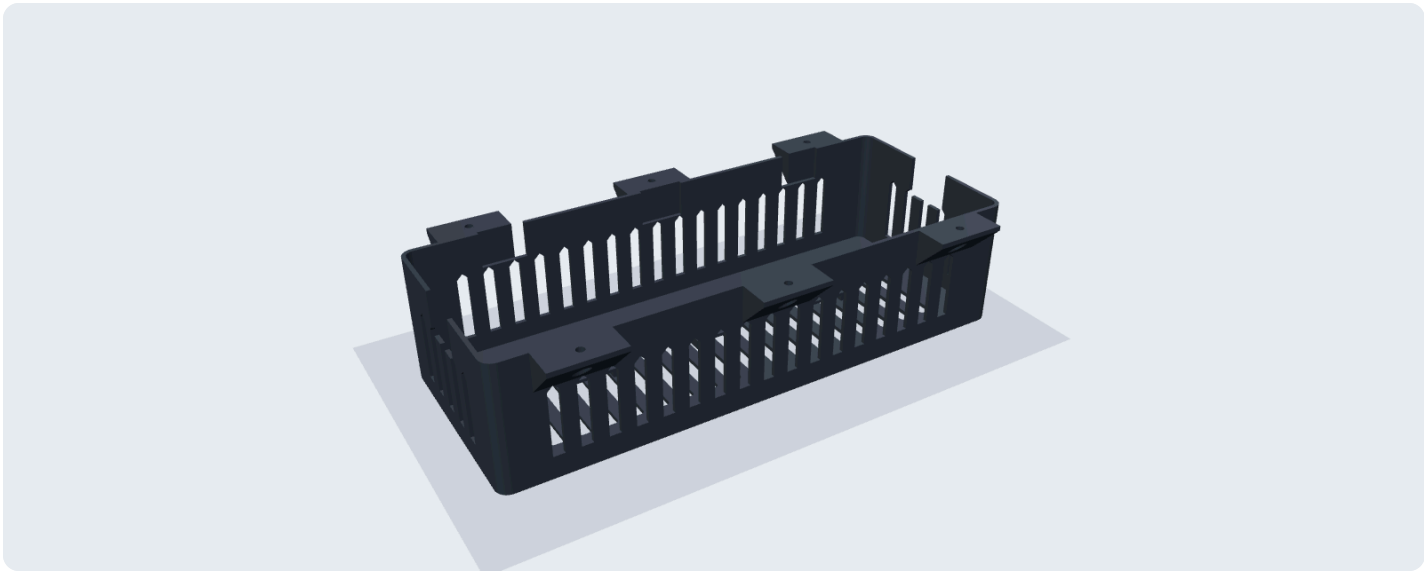
| PART                | SIZE, MM        | PRINT TIME            | FILAMENT |
|---------------------|-----------------|-----------------------|----------|
| C1 base             | 300 × 57 × 24   | 1 h 56 min            | 79.6 g   |
| C2 base, side exits | 300 × 57 × 24   | 1 h 57 min            | 78.3 g   |
| C3 lid              | 300 × 60 × 10.5 | 1 h 07 min            | 46.9 g   |
| C4 joiner           | 40 × 29 × 10    | 17 min                | 3.3 g    |
| Total (base + lid)  |                 | 3 h 03 min per 300 mm | 126.5 g  |



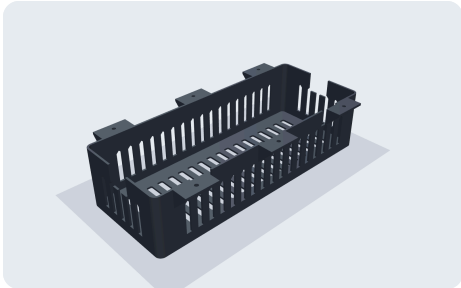
# Under-Desk Basket

OVERNIGHT

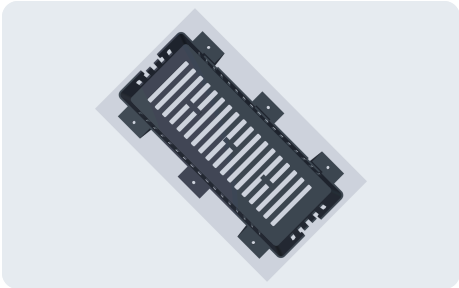
A vented tray that screws under the desk and holds the power strip, the router's power brick and the extra cable length, so nothing lies on the floor.



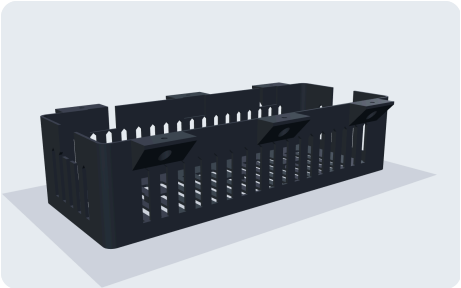
D1 basket



From the back: cord notches



From above: vented floor



Screw brackets

## What it does

- Inside 325 × 125 × 77 mm, fits most 6-outlet power strips.
- Six screw points on solid 45° brackets (no supports needed); heads sit in recesses under the flange.
- Cord notches in the back wall and both ends; three strap loops on the floor for the strip.
- Vented floor and walls for the brick's heat.

## How to use it

1. Choose a spot under the desk and hold the basket up to mark the six holes.
2. Use screws at least 3 mm shorter than the desk is thick (for an 18 mm desk, 15 mm screws). Drill small pilot holes first.
3. Put the power strip and brick inside and strap the strip down through the floor loops.
4. Run the cords out through the notches in the back wall and the ends.

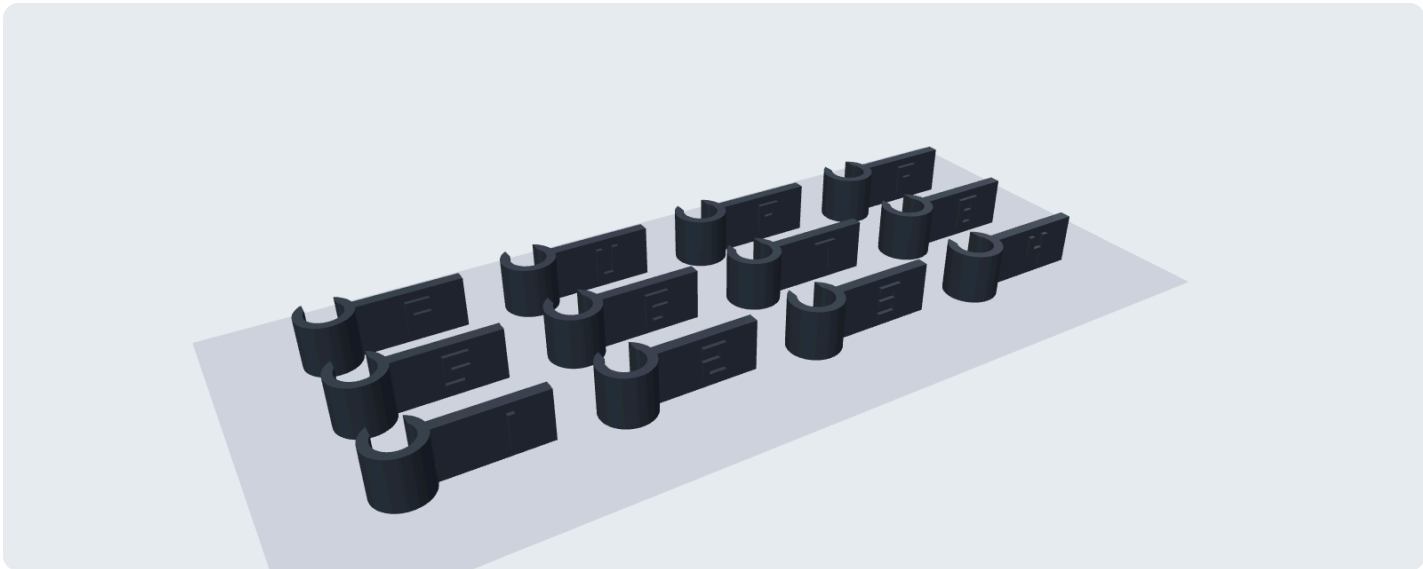
| PART      | SIZE, MM       | PRINT TIME | FILAMENT |
|-----------|----------------|------------|----------|
| D1 basket | 330 × 170 × 80 | 8 h 27 min | 236.6 g  |



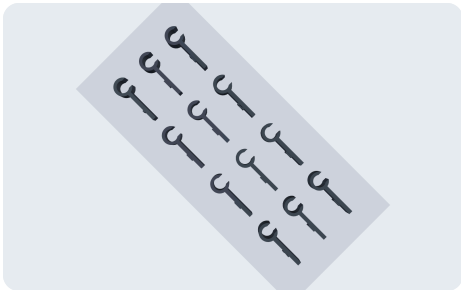
# Tags and Holders

QUICK

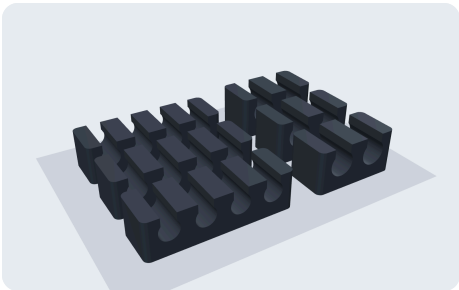
Small helpers: numbered tags that clip onto each cable so you know which port it goes to, and stick-on holders for the far end of a run.



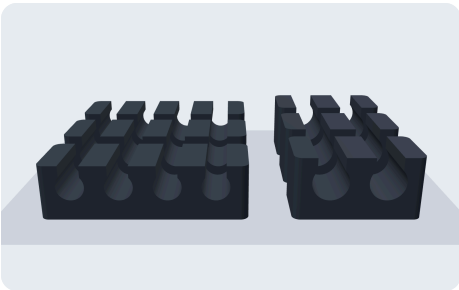
E1 tags



Tags from above



E2 stick-on holders



Holder channels

## What it does

- Twelve tags: 1–8, F, U and two P. They clip onto 5.5–6.5 mm cables with the number facing out.
- Six holders: three with four 6 mm channels for network cables and three with two 8 mm channels for power cords. Stick them on with double-sided tape.

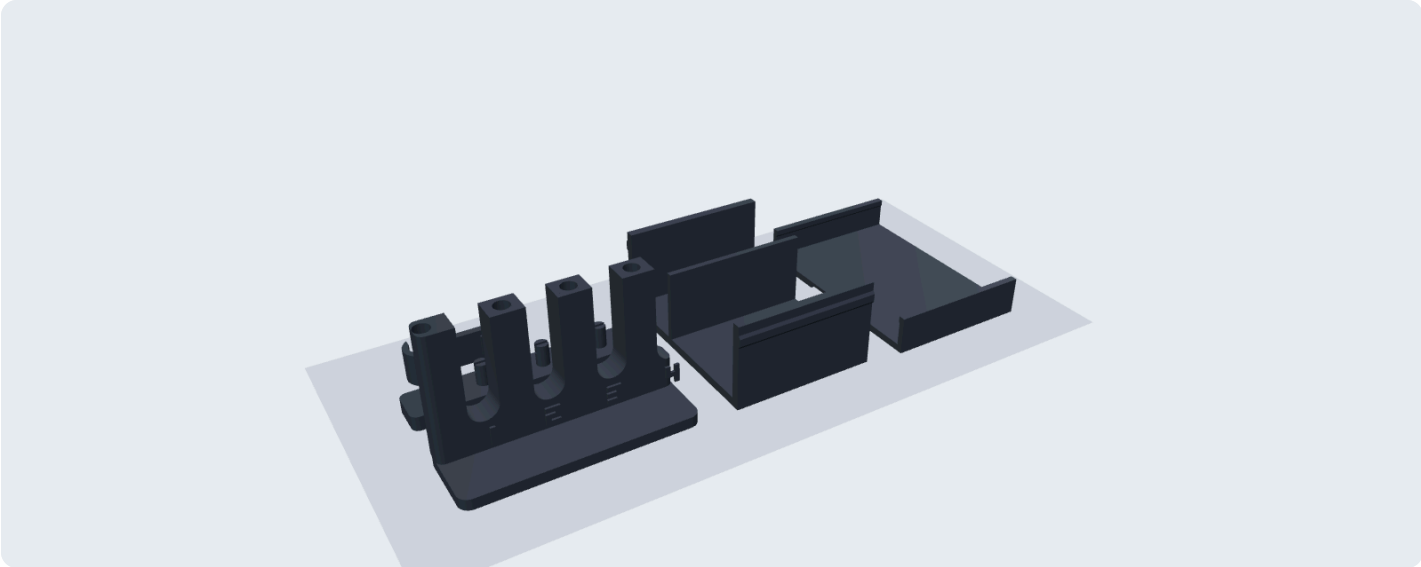
## How to use it

1. Clip a tag onto each cable near both ends, number facing out.
2. Stick holders on with double-sided tape on a clean, dry surface; press for 30 seconds.
3. Use the 6 mm holders for network cables and the 8 mm holders for power cords.

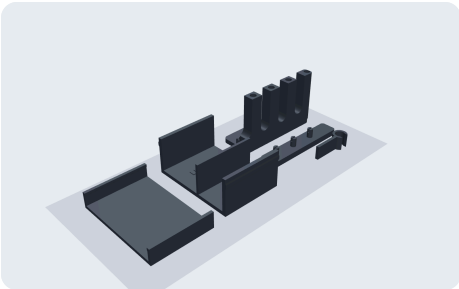
| PART       | SIZE, MM     | PRINT TIME | FILAMENT |
|------------|--------------|------------|----------|
| E1 tags    | 129 × 45 × 9 | 30 min     | 8.3 g    |
| E2 holders | 91 × 60 × 16 | 1 h 05 min | 23.3 g   |
| Total      |              | 1 h 35 min | 31.5 g   |



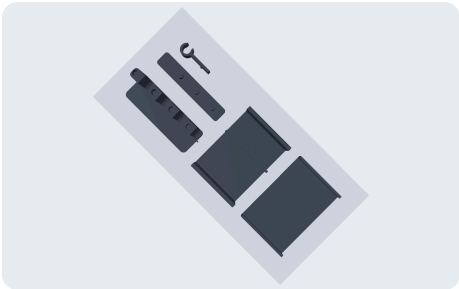
One small plate with a sample of each tricky fit. Print it before a long job and check it against the real router and cables.



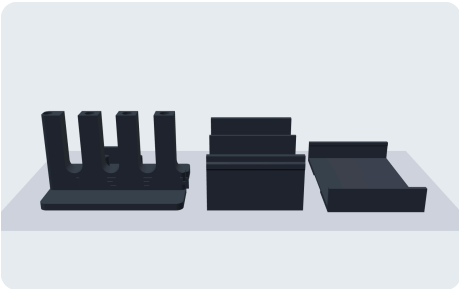
F1 test plate



From behind



From above



Front: slots 1-3

### What it does

- Three comb slots at the spacing of ports 1–3, with a short lock bar.
- A 40 mm raceway base and lid, to feel the snap.
- One cable tag, to try on your thickest cable.

### How to use it

1. Print it (about 1¼ hours).
2. Hold the comb section about 4.5 cm in front of ports 1–3. The cables should run straight into slots 1, 2 and 3.
3. Press the small lock bar on, then the raceway lid on and off a few times: it should click and stay.
4. Clip the tag onto your thickest cable.
5. Tell Claude what you found. If the slots are off, the designs get adjusted before the long prints.

| PART          | SIZE, MM      | PRINT TIME | FILAMENT |
|---------------|---------------|------------|----------|
| F1 test plate | 160 × 60 × 37 | 1 h 14 min | 29.0 g   |

# Edge cases

| SITUATION                             | HOW THE KIT HANDLES IT                                                                                                                                                                                                                                                                         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Port positions                        | MikroTik publishes no port drawing, so slot positions come from the standard 15.88 mm jack pitch and product photos. Slots are 7.6 mm wide and sit 45 mm in front of the ports, so a few millimetres of error only angles a cable slightly. Plate F tests three slots against ports 1–3 first. |
| Different cable sizes                 | Slots take Cat5e (5 mm) through Cat6 (6–6.5 mm), flat cables on edge, DAC and USB cables. Tags clip onto 5.5–6.5 mm cables. Holders come in 6 mm and 8 mm.                                                                                                                                     |
| Adding or removing a cable later      | Every slot, lane and holder opens from the top, so no plug has to pass through a hole. Lift the lock bar or lid, drop the cable in and close it.                                                                                                                                               |
| Router heat                           | The RB5009 cools through its case. The dock touches it only on thin rails and corner posts, leaves a 3.5 mm air gap with vent windows underneath, and keeps the top and fins open. PLA softens around 55–60 °C; if the case gets too hot to hold comfortably, reprint the dock in PETG.        |
| Power brick heat                      | The brick bay and the basket have open tops and vented walls, and nothing wraps the brick tightly.                                                                                                                                                                                             |
| Power next to data                    | The DC cord has its own lane in the dock, the raceway has a walled power lane, and the mains cord leaves the brick bay at the back.                                                                                                                                                            |
| Buttons, lights and the side terminal | Reset button, LEDs, USB and SFP+ stay reachable. The corner posts are 16 mm long, so the 2-pin power terminal and ground screw on the left end stay uncovered. The router lifts out without tools.                                                                                             |
| Cable bending                         | Cables run straight from the port to the comb (Cat6 wants a bend radius of 25 mm or more; fiber 30 mm). Strap loops sit after the comb, not at the port.                                                                                                                                       |
| Screws                                | Countersunk holes for #8 or M4 wood screws, or tape on any flat back. The basket screws up into the desk with the heads recessed. Use screws at least 3 mm shorter than the desk is thick.                                                                                                     |
| Printing                              | Every part prints flat, needs no supports and fits the 390 mm bed. The raceway lid prints upside down; lock-bar pins print pointing up.                                                                                                                                                        |
| Snap fits                             | The raceway lid flexes about 0.6 mm to click on (about 2 % strain, safe for PLA) and has a finger notch to pry it off. Lock-bar pins have a slit so they press in without cracking the comb.                                                                                                   |

## Printing a part

1. Tell Claude which letters you want, for example "F, then A overnight", and any changes.
2. Claude slices it with the locked profile, runs the safety check and uploads it, then the print starts only after you confirm at the printer.
3. A monitor watches temperatures and progress during the print and stops it on a fault.